Karger contraction always finds a minimum cut on a cycle

Disproof of Conjecture 00000007841

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Abstract

For every simple cycle C_n, $n \geq 3$, Karger's uniform-edge contraction algorithm returns some minimum cut with probability one. Therefore the claimed bound $2(1 - \varepsilon(n))/n^2$, with $\varepsilon(n) = O(1/n)$, cannot be sharp on cycles: it tends to zero and its gap from the actual success probability tends to one. Lean verifies actual simple input graphs, genuine multigraph contractions, all histories and normalized sequential edge-sampling probabilities.

Which success event the source defines

The source defines $p(G)$ as the probability that the algorithm outputs *any* minimum cut. It asserts that its displayed lower bound is tight on cycles; the Chinese text explicitly claims equality there. We refute this sharpness clause for arbitrarily large simple connected cycles. The separate assertion about the infimum ratio $\rho(n)$ is not assessed here.

Karger's algorithm repeatedly chooses a current non-loop edge uniformly, merges its endpoints and deletes loops, retaining parallel edges. At two supervertices it returns their bipartition, pulled back to the original vertices. This is the ordinary contraction algorithm of [2, Section 2].

An explicit, genuine contraction of a cycle

Represent the vertices and edge labels of C_m by $\mathbb{Z}/m\mathbb{Z}$, with edge i joining i to $i+1$. For $m \geq 3$ this is the usual simple cycle. For $m = 2$ the same labelled model has two parallel edges; both must be retained.

Given an edge e of C_m, $m \geq 3$, rotate it to edge zero by $x \mapsto x - e$. With representatives $0 \leq x < m$, use the quotient map

$$q(x) = \max(x - 1, 0) \pmod{m - 1}.$$

Only vertices 0 and 1 are identified. Edge zero becomes a loop and is deleted. Each remaining edge $i = 1, \ldots, m - 1$ maps to edge $i - 1$ of C_{m-1}, with the correct two endpoints. Thus the remaining edges correspond bijectively, including their multiplicities, to the next cycle multigraph.

Every contraction history succeeds

For a Boolean vertex bipartition A, its actual crossing-edge count is

$$c_m(A) = \sum_{i \in \mathbb{Z}/m\mathbb{Z}} \mathbf{1}_{A(i) \neq A(i+1)}.$$

Around a cycle the number of transitions from zero to one equals the number from one to zero. Consequently $c_m(A)$ is even. A nonconstant bipartition has a crossing edge, since equality at every consecutive pair would make the assignment constant. Every nontrivial cut therefore has size at least two.

At order two, the final bipartition has two crossing parallel edges. Pulling a proper cut back through a surjective contraction preserves properness. It also preserves its crossing-edge count: the selected edge has both endpoints in one fiber and contributes zero, while every other edge corresponds bijectively to a target edge. Induction backwards through the entire history shows that every returned cut of the original cycle is proper and has size two. It is therefore an actual minimum cut, for every possible history.

At current order m there are exactly m non-loop edge labels, so every one is selected with probability $1/m$. The successive choices have orders $n, n-1, \ldots, 3$. The full history set has size

$$|\mathcal{H}_n| = \prod_{m=3}^{n} m = \frac{n!}{2},$$

and each history has its actual sequential probability

$$w(h) = \prod_{m=3}^{n} \frac{1}{m} = \frac{2}{n!}.$$

These weights are nonnegative and sum to one. The success event is the assertion that the returned proper cut is no larger than *every* other proper cut. Since all histories satisfy this event,

$$p(C_n) = \sum_{h \in \mathcal{H}_n} w(h) \mathbf{1}_{\mathrm{success}(h)} = 1 \qquad (n \geq 3).$$

This probability is derived from the actual sampling rule and actual returned partitions; it is not assigned the value one by definition.

The claimed cycle sharpness is impossible

If $\varepsilon(n) = O(1/n)$, then $\varepsilon(n) \to 0$. In fact under the weaker condition $\varepsilon(n) \to 0$, the proposed expression satisfies

$$b_n := \frac{2(1-\varepsilon(n))}{n^2} \longrightarrow 0, \qquad p(C_n) - b_n \longrightarrow 1.$$

It cannot equal the actual success probability on cycles, even eventually, and it cannot asymptotically match it. Lean also gives a quantitative contradiction: for every eventual bound $|\varepsilon(n)| \le C/n$ and every cutoff, some larger cycle has $b_n < p(C_n)$.

Theorem 1. *The lower bound stated in Conjecture 00000007841 is not sharp on simple cycles, whether sharpness means its asserted exact equality or asymptotic agreement with the actual success probability.*

Proof. All simple cycles have success probability one, whereas the displayed expression tends to zero. Their difference tends to one, not zero. □

The familiar contraction estimate concerns survival of one *prescribed* minimum cut [2, Theorem 2.1]. On a cycle there are many different minimum cuts, and every returned cut is minimum. The source defines success as returning any one of them. This distinction explains the error. The disproof targets its cycle-sharpness conjunct; no claim about the actual extremizers or limit of $\rho(n)$ is needed.

Bridges to the actual graph and algorithm

The Lean input is Mathlib's actual simple connected `cycleGraph`. Theorems identify its full undirected edge set with the cyclic edge labels and prove equality between its genuine crossing-edge count and the displayed sum. Vertex and edge counts are checked for every input order at least three.

The contraction map has exactly the selected edge's two endpoints as its only nonsingleton fiber. Theorems verify surjectivity, deletion of exactly that loop and preservation of every other edge's endpoints and identity. Intermediate order-two states use the correct two-edge multigraph, not a simplified single edge. The returned cut is recursively pulled back through these same quotient maps.

Formal conclusions and reproducibility

The Lean type `History k` starts at order $k+2$. The simple input `inputGraph n` of order $n+3$ therefore uses `History (n+1)`. The theorem `cycle_counterexample` bundles the actual simple connected input, its counts, success probability one and minimum output cut size two.

The event `Success` compares an actual proper output partition with every proper partition. The function `pathWeight` multiplies the true successive $1/m$ factors. Its nonnegativity and total mass one are proved explicitly. The definition `actualSuccessProbability` sums these actual weights over that event. Graph-level correctness is proved by `actual_graph_success`.

The probability-one theorem is named `actual_cycle_success_probability`. The final limit theorems prove that the gap tends to one and cannot tend to zero, alongside the quantitative arbitrary-cutoff contradiction.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The complete project is built afresh and checked with warnings treated as errors. Printed main-theorem axiom dependencies contain only standard foundational axioms; no proof placeholders, custom axioms or external computation oracles are used.

The supplementary `verify.py` independently enumerates contractions by original edge identities, deleting loops and retaining parallel edges. It checks C_3 through C_9 exactly, and complete graphs up to K_7, including the nontrivial result $p(K_4) = 4/5$. This is supplementary finite verification; Lean proves the full cycle family. Run `python verify.py` and `tectonic main.tex`.

The exact source, provenance, build logs, artifact hashes and PDF checks are included. An agent developed the proof and supplied a self-review. The parent separately reviewed the complete proof and final artifacts. No external independent review or novelty is claimed for these standard contraction facts.

References

[1] The Last Math Competition, Conjecture 00000007841. Exact bilingual source and provenance included.

[2] D. R. Karger and C. Stein, *An $\widetilde{O}(n^2)$ Algorithm for Minimum Cuts*, Section 2 and Theorem 2.1. Author-hosted paper. The reference supplies the standard algorithm and prescribed-cut context; the complete cycle argument used in this disproof is given above.